INCH-POUND
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PERFORMANCE SPECIFICATION

TESTER, PRESSURIZED CABIN LEAKAGE, LARGE AIRCRAFT, TRAILER MOUNTED

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This trailer mounted large aircraft pressurized cabin leakage tester specification covers only one type of tester designated as AF/M32T-4. This tester is a self contained, enclosed, four wheel, pneumatic tire, towable, trailer mounted, diesel engine powerplant, cabin leakage tester capable of 10 - 1000 cubic feet per minute (cfm) at 2 - 12.5 pounds per square inch gage (psig).

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications and standards. The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation (see 6.2).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: SA-ALC/TILDD, Kelly AFB TX 78241-5916 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 4920

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SPECIFICATIONS

DEPARTMENT OF DEFENSE

MS24397	-	Reducer, Boss, Precision Type
MS28049	-	Indicator, Vertical Speed, 3 Inch
MS33660	-	Tubing End, Hose Connections, Standard Dimension for

STANDARDS

FEDERAL

FED-STD-595	-	Colors used in Government Procurement

MILITARY

MIL-STD-209	-	Lifting and Tie-down Provisions
MIL-STD-461	-	Requirements for the Control of Electromagnetic
MIL-STD-1366	-	Transportability Criteria

AIR FORCE - NAVY AERONAUTICAL

AFI 32-7086	-	Hazardous Material Management
AN832	-	Union, Flared Tube, 3/8 Bulkhead and Universal.
AN893	-	Bushing-Screw Thread Reducer
AN924	-	Nut, Tube, Bulkhead and Universal Fitting.
AN929	-	Cap Assembly, Tube, Pressure Seal.
AN6270	-	Hose Assembly -- Detachable Swivel Fitting, Low Pressure.

(Unless otherwise indicated, copies of the above specifications and standards are available from the Standardization document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094).

2.3 Non-Government publications. The following document form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD are those listed in the issue of the DoDISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS are the same issues of the documents cited in the solicitation (see 6.2).

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI-B40.1-85	-	Gauges - Pressure Indicating Dial Type – Elastic Element (DoD adopted)

(Application for copies should be addressed to the American National Standard Institute (ANSI), 11 W. 42ND Street, 13th Floor, New York, NY 10036)

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2.4 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Components. The tester shall be manufactured using the latest technology available. However, at a minimum, it should consist of the following major components:

a. Diesel Engine System (see 3.7)

b. Canopy Seal System (see 3.8)

c. Pressurization System (see 3.9)

d. Instrument and Control Panel (see 3.10)

e. Housing (see 3.11)

f. Trailer (see 3.12)

g. Accessory Equipment (see 3.13)

3.2 Materials.

3.2.1 Recycled, recovered, or environmentally preferable materials. Recycled, recovered, or environmentally preferable materials should be used to the maximum extent possible provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

3.3 Processes. The processes used for the testers shall not rely on hazardous or toxic materials, or Class I Ozone Depleting Chemicals (ODCs), in accordance with AFI 32-7086.

3.4 Protective treatment. When materials used in the construction of the tester are subject to deterioration when exposed to climatic and environmental conditions, likely to occur during service usage, they shall be protected against such deterioration in a manner that will in no way prevent compliance with the performance requirements of this specification. The use of any protective coating that will crack, chip, or scale with age or extremes of climatic and environmental conditions shall be avoided.

3.5 Performance requirements. The tester shall provide a range of 10 through 1000 cfm; standard airflow is 0.765 through 76.5 pounds per minute (ppm) with a pressure range of 2 through 12.5 psig. The canopy seal pressure system shall provide 1 cfm standard air at 65 psi.

3.5.1 Safety devices. Shields and caution notes shall be furnished where necessary and maximum use shall be made of applicable safety devices.

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3.5.1.1 Danger zone. A red line zone shall be marked on the air temperature gauge face beginning at 120°F.

3.5.1.2 Relief device. A relief device shall be incorporated to prevent over pressurization of aircraft under test.

3.5.2 Maintenance and servicing. The tester shall be constructed so that lubrication, adjustments, repairs, and overhaul can be easily accomplished by personnel without the use of special tools. All components shall be selected and located for ease of operation, servicing, inspections, and maintenance shall be able to service and visually inspect without removing panels or components.

3.5.3 Accessibility of controls. All operating and service controls shall be readily accessible and easily operated by personnel wearing Air Force issue type leather gloves.

3.5.4 Noise limit. Acoustical noise limits shall be those that are acceptable and current for industrial application and shall be no more than 95 dbA (decibels "A" level). These limits shall not be exceeded at ten feet from the periphery of the tester. At the operator position, the acoustical noise limits shall be no more than 100 dbA (decibels "A" level).

3.5.5 Reliability. The tester shall have a minimum mean time between failure (MTBF) of 500 hours.

3.5.6 Maintainability. The tester shall be so constructed to enhance accessibility for troubleshooting and repair. No special tools will be required unless specifically authorized by the procuring activity. The mean time to repair (MTTR) shall not exceed two hours including troubleshooting, removal and replacement of defective components and operational checkout of the tester. The maximum downtime for a single failure shall not exceed four hours.

3.5.6.1 Preventive maintenance. The tester shall have a mean time between preventive maintenance (MTBPM) of 100 hours.

3.6 Performance. The tester shall be capable of satisfactory operation after exposure to the following conditions:

a. Storage temperatures ranging from -65°F to 165°F.

b. Operating temperature from -40°F to 125°F.

c. Rainfall minimum rate of 4 inches per hour.

d. Altitude ranging from sea level to 6,000 feet above sea level.

3.7 Diesel engine system. Unless otherwise specified, a diesel engine and all accessories shall be selected and provided by the tester contractor in accordance with Appendix A.

3.8 Canopy seal system. The canopy seal system shall be capable of continuously providing 1 cfm standard oil-free air at 65 psig for pressurizing aircraft canopy seals. The output of the compressor shall be controlled by an automatic pressure regulator with manual adjustment to maintain any pressure setting

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throughout a range of 5 psi to 60 psi. A safety device shall be provided to limit maximum pressure to 65 psi. A pressure gage, 0 to 100 psi scale (+2% of full scale), shall be provided to indicate canopy seal pressure. The canopy seal pressure outlet on the tester shall have an AN832-6D fitting, and AN924-6D nut with an AN929-6D dust cap.

3.8.1 Air compressor. The air compressor for the canopy seal system shall be the type that is currently acceptable for industrial use. A maximum pressure relief valve set at 70 psig shall be provided in the system.

3.9 Pressurization system. The pressurization system shall consist of the following:

3.9.1 Blower. A positive displacement blower with inter-cooling shall be provided to supply 1000 cfm standard air (76.5 ppm) at 12.5 psi measured at the discharge end of the outlet hose. The bearings used in the blower shall be easily replaceable.

3.9.1.1 Drive shaft clearance. Means shall be provided for sealing the external and internal drive shaft clearance holes in the end plates to prevent leakage. The drive shaft shall be covered to prevent accidental contact with the drive shaft by personnel.

3.9.1.2 Installation and removal. Alignment of blower to engine shall not require special tools for installation or removal.

3.9.2 Intake silencer. The blower intake shall be equipped with an air-intake/cleaner/silencer device that does not protrude more than 3 inches from the tester exterior surface.

3.9.3 Blower outlet silencer. A silencer shall be provided to reduce noise and dampen the pulsation of the blower discharge air.

3.9.4 Aftercooler. An air-to-air aftercooler capable of providing the required cooling and volume of air shall be provided. The aftercooler shall have sufficient capacity to cool blower discharged air to within 25°F of existing ambient temperature while delivering maximum flow and pressure.

3.9.4.1 Aftercooler fan. An aftercooler fan shall be provided to supply the necessary cooling air to the aftercooler.

3.9.4.2 Aftercooler fan inlet. The aftercooler fan inlet shall be equipped with adjustable louvers, which lock in any degree of opening.

3.9.4.3 Aftercooler fan exhaust. The aftercooler fan exhaust shall extend to the outside of the tester housing and be protected to prevent accidental contact by personnel.

3.9.5 Pressure regulating device. A pressure-regulating device shall be provided to automatically control the pressure in the aircraft cabin being pressurized, and shall be installed between the aftercooler and flowmeter. Pressure regulating valve(s) shall be capable of limiting the rate of pressure change to 2.47 psi per minute, equivalent to 5000 feet per minute (FPM) on the rate of climb indicator.

3.9.5.1 Pressure regulating valve. The pressure regulating valve shall be capable of bypassing the blower output limiting the discharge pressure to 1.5 psi of pressure setting.

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3.9.5.2 Valve control system. The valve control system shall be manually adjustable and will remain fixed at any pressure set within the range of 2 through 12.5 psi for all flow and temperature conditions specified herein. The manual adjustment for the valve control system shall be located on the instrument panel.

3.9.5.3 System dumping. A means shall be provided to dump the system at a minimum rate of 80 cfm with the unit running.

3.9.6 Air outlet connection. A 4 inch hose connection in accordance with MS33660 shall be provided at the flowmeter discharge and shall extend 6 inches beyond the test specimen housing for hose connection. The outlet shall be rigidly supported to prevent damage to the flowmeter in the event the protruding section of the outlet is severely damaged, twisted, or sheared off. A protective cap shall be provided to protect all air outlets.

3.9.7 Air ducts. The air ducts and passages between the components specified herein shall be a minimum of elbows, and restrictions that would create an excessive pressure drop through the system. Hoses of proper temperature and pressure ratings are allowed between the blower discharge and heat exchanger inlet.

3.10 Instrument control panel. The tester shall have a control panel on which all necessary instruments, gages, switches, valves, and controls shall be mounted. The panel and its accessories shall be readily accessible to operating personnel, and all items shall be so mounted that they may be easily removed for replacement or servicing. Instruments subject to errors from vibration shall be shock mounted. All items mounted on the panel shall be permanently and legibly identified with nameplates. A 4 by 6 inch, engraved, permanent-type data plate showing the pneumatic circuit shall be securely fastened on or near the control panel. Voltages greater than 12V shall not be used on the control panel circuitry. The panel shall be located at sufficient height above the ground to accommodate a standing operator and inclined at a minimum 10° angle.

3.10.1 Flowmeter. This flowmeter shall measure the airflow to the aircraft being pressurized. The flowmeter shall provide a range of 10 through 1000 cfm standard air at pressures of 2.0 through 14.0 psi. A multi-stage turbine type meter with LED shall be provided with accuracy to 2 percent minimum of scale. Protective covering shall be provided over the flowmeter to protect the operator from injury in case the flowmeter should break or explode.

3.10.2 Temperature correction curve. A permanent data plate that is legible and waterproof, identifying the pressure-temperature correction curve, shall be provided and mounted on the panel adjacent to the flowmeter for correcting observed airflow readings to ambient conditions. Pressure lines shall be shown and marked to correspond with pressure lines on the flowmeter.

3.10.3 Conversion chart. A permanent data plate that is a legible and waterproof, identifying conversion chart, shall be provided and located on the panel for converting the airflow from cubic feet per minute to pounds per minute.

3.10.4 Pressure equalization indicator. A rate of climb indicator per MS28049 shall be furnished to indicate pressure equalization of the tester output and the enclosure being pressurized. The dial face shall indicate decreasing and increasing pressurization instead of up and down. The indicator shall be redlined beyond the 5,000-foot mark, both increasing and decreasing. Some means shall be provided to

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prevent damage to the instrument should the rate of increasing or decreasing pressure exceed the redlined values. A permanent caution plate containing the following information shall be securely attached to the panel immediately below the indicator.

CAUTION -- DO NOT INCREASE OR DECREASE CABIN
PRESSURIZATION AT THE RED-LINED RATES

3.10.5 Cabin pressure gage. The corrosion-resistant gage shall be accurate to within 1 percent or less of full scale and shall conform to ANSI B40.1 except as specified herein. The gage shall provide a range of from 0 to 15 psi and shall incorporate a means to restore the accuracy of the gage under all service conditions. The gage shall be suitable for operation from -25°F to 160°F. The gage shall be of the recess type provided with a mounting flange at the front of the gage. The mounting flange and dial shall not extend more than 0.625 inch beyond the external surface of the panel. The pointer shall be adjustable without removing the gage from the panel.

3.10.6 Discharge air temperature indicator. A remote-reading, dial-type temperature indicator having a scale range of from -25°F to approximately 200°F and accurate within ± 2°F throughout the range shall be furnished. The temperature indicator shall measure the temperature of the pressurized air on the discharge side of the flowmeter and shall be marked by a red line above 120°F. A permanent caution plate containing the following information shall be securely attached to the panel immediately below the indicator.

CAUTION -- AIR TO CABIN SHOULD NOT EXCEED 120°F.

3.10.7 Pressure gages. The corrosion-resistant gages specified herein shall be included on the panel. Each pressure gage shall incorporate a safety blowout device that shall relieve the gage case pressure due to gage failure. The dial and pointer shall be separated from the gage movement to prevent rupture through the front of the gage in the event of gage failure. The pointer shall be adjustable without removing the gage from the panel. All gages shall be anti-vibration mounted.

3.10.7.1 Gages connection. Pressure gages shall be connected with flexible AN or MS type hose, or commercial equivalent hose and be of sufficient length to permit removal and installation from the front of the panel.

3.10.7.2 Gage test fitting. The following three items when assembled compose the fitting to be installed on the control panel: AN832-4D nipple, AN924-4D nut, and AN929-4 cap.

3.10.7.3 Fittings and Tubing. For interoperability the following three fittings shall be mounted on the control panel adjacent to the appropriate gages, and the panel shall be marked to indicate that the fittings are gage test connections. The outlets from the tester include:

a. Cabin Pressure. Fitting AN832-8D with AN 929-8 cap assembly

b. Canopy Pressure. Fitting AN832-6D with AN 929-6 cap assembly

c. Static Pressure. Fitting AN832-8D with AN929-8 cap assembly

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d. Static System. The tubing size of 0.25 inch OD may be used for internal connection lines for static systems only.

3.10.8 Blower pressure gage. A pressure gage having a minimum dial size of 3 inches, a range of 0 to 20 psi, and minimum accuracy of 1 percent of full scale shall be furnished to indicate the blower pressure at the inlet side of the flowmeter.

3.10.9 Canopy seal pressure gage. A pressure gage with a scale range of 0 to 100 psi, accurate to within 2 percent of full scale, shall be provided in the canopy seal system.

3.10.10 Hourmeter. A hourmeter having a range of from 0 to 9,999 hours shall be provided to record the hours of operation of the blower motor.

3.11 Housing. The tester components shall be completely enclosed in a weather-resistant housing. The housing shall utilize doors and adjustable louvers as required. The housing shall be capable of supporting a 200 pound weight distributed evenly over an area of 4.5 by 5 inches at any location on top of the housing. The housing shall be adequately braced to prevent distortion or excessive vibration during handling and operation of the tester or handling of the housing while being removed from the tester.

3.11.1 Doors. The housing shall be provided with sufficient access and ventilating doors and adjustable louvers to permit access for operating and servicing, and to ventilate the tester. The doors and louvers in the housing shall be free from binding and easily operated with one hand. Provisions shall be incorporated to secure the doors rigidly in the fully open or fully closed position. An access port in the housing must be provided to allow removal of temperature sensor from the air outlet.

3.11.2 Handles. The housing shall be easily removable by actuating not more than 12 quick-acting disconnecting fasteners and lifting or sliding the entire housing off the remainder of the tester. Handles or lifting eyes shall be provided for this purpose.

3.12 Trailer. The components of the tester shall be mounted and assembled on a trailer having mobility requirements as specified herein.

3.12.1 Tires. The four trailer tires shall be adequate sized industrial, pneumatic tires.

3.12.2 Trailer bed. The trailer bed shall be constructed of a commercially acceptable material. The ratio of width divided by length shall not exceed three-fourths.

3.12.3 Parking brake. The trailer parking brake(s) shall hold the tester in a fixed position during all operating conditions and positions specified herein. The brake(s) shall be of the internal expanding type capable of locking both rear wheels using one handle.

3.12.3.1 Marking of brake control device. The parking brake control device shall be appropriately marked with operating instructions for release and application of the brakes. The marking shall be 0.25 inch black letters and shall be near the parking brake control handle.

3.12.4 Springs. Springs shall be furnished on the trailer to provide good riding qualities, thereby precluding damage to the mounted equipment.

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3.12.5 Steering. Adequate means for lubrication of the steering linkage shall be provided. The cramping angle shall be a minimum of 40°.

3.12.6 Towbar assembly. The trailer shall be furnished with a towbar assembly which shall be stored in the up position while the tester is in use or parked.

3.12.7 Mobility. The tester shall be capable of executing full turns at a speed of 10 mile per hour without skidding or tipping over on dry concrete or macadam.

3.12.8 Lifting and tie down fittings. The trailer shall be equipped with a minimum of four lifting and tie-down fittings as specified in MIL-STD-1366D (paragraph 5.7) and MIL-STD-209J. Slots for forklift tines will be provided.

3.12.9 Road clearance. The fully loaded trailer shall have a minimum 6 inch road clearance.

3.13 Accessory equipment. For interoperability the following accessories shall be furnished:

a. One AN6270-8-480 hose assembly. (Dust plugs, securely attached to the hose with metal chain, shall be provided for each end fitting of hose.)

b. One AN6270-600 hose assembly. (Dust plugs, securely attached to the hose with metal chain, shall be provided for each end fitting of hose.)

c. One AN893-12D bushing to adapt the 0.5 inch hose to the 0.375 inch static fitting on the aircraft.

d. One AN893-2D bushing to adapt the 0.375 inch hose to the 0.25 inch canopy seal fitting on the aircraft.

e. One 40 foot length of 4 inch inside diameter (ID) flexible hose, National Fire Hose, Part Number 44-250 (CAGE 30951), or equivalent, and the necessary clamps and adapters to secure the hose to the outlet specified in 3.9.6 and to the aircraft under test (e.g. B-1, KC135, EC135). Aircraft to be tested will be specified in the procurement contract using this performance specification.

f. One stethoscope for detecting cabin leakage (airflow) through the skin of the aircraft. The stethoscope shall be suitable for use at -25°F and shall be of rugged construction to withstand maintenance personnel service usage.

3.13.1 Accessory box. An accessory box having separate compartments shall be provided within the housing for storage of the accessory fittings, adapters, clamps and gaskets specified in 3.13 a, b, c and d. And the second compartment shall contain the stethoscope specified in 3.13.f. The box shall have a hinged cover with a hasp for possible padlocking and for holding the cover closed.

3.13.2 Pressurization hose storage device. A rack, tray, or compartment shall be provided within the tester housing to store the pressurization hose specified in 3.13.f. In addition, hooks shall be provided on the exterior of the housing to permit temporary storage of all hoses without disconnecting them from the outlets. The hooks shall be designed to retract against the housing when not in use.

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3.13.3 Handbook compartment. A handbook compartment shall be provided and permanently mounted within the tester housing. This compartment interior shall be a minimum size of 10 by 12 by 4 inches.

3.14 Interchangeability. All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable. The drawing number and part numbers shall be structured in a manner to identify the item.

3.15 Dimensions. The length and width of the tester and housing shall be not less than the trailer bed and shall not overhang the outline of the wheels at any point. The overall height of the tester shall be held to the minimum. Unit shall be designed for shipment and size and weight shall be in accordance with current industry requirements.

3.16 Cleaning, plating, and chemical films. Cleaning, plating and chemical films shall be in accordance with acceptable, current industry practices as specified herein. The color of the tester and the inner surfaces of exterior panels or components will be dark green, color 24052, Film Designation LG, per FED STD 595.

4. VERIFICATION

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure prescribed requirements.

4.1.1 Classification of tests. The inspection and testing of test specimens shall be classified as Individual Acceptance Test. The number of units to be tested that are produced under contract shall be specified as part of the contract.

4.2 Individual acceptance test. The identified units shall be subjected to the following tests as described under 4.4.

a. Examination of product (see 4.4.1).

b. Operational test (see 4.4.2).

c. Calibration test (see 4.4.3).

d. Environmental test (see 4.4.4).

4.3 Test conditions.

4.3.1 Standard atmospheric conditions. When the pressure and temperature existing at the time of the test are not specified definitely, it is understood that the test is to be made at atmospheric pressure, approximately 29.92 inches Hg, and at room temperature, approximately 73°F. All airflow readings shall be reduced to standard air readings of 0.07651 PCF at 60°F.

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4.3.2 Apparatus and instruments. Apparatus and instruments used in conjunction with testing specified herein shall be of laboratory precision type and shall be carefully checked to ascertain that they have been installed in a satisfactory manner for durability and accurate operation and shall be calibrated at intervals properly spaced to assure laboratory efficiency.

4.3.3 Airflow. The airflow shall be measured by means of the most accurate, scientifically recognized techniques, instruments and methods and shall be acceptable when certified.

4.3.4 Test periods. Unless otherwise specified, any test run shorter than 1 hour shall not be counted as test time.

4.3.5 Preliminary run-in and calibration. Any preliminary run-in of the tester as recommended by the manufacturer shall be performed by the manufacturer prior to submission of test specimens for testing. A certified log of all preliminary running time and calibration data shall be submitted to the procuring activity.

4.4 Test methods. The following test methods shall be used for Individual Acceptance Test product assurance.

4.4.1 Examination of product. The tester shall be inspected to determine compliance with requirements specified herein with respect to parts, materials, workmanship, dimensions, and marking.

4.4.2 Operational test. With hoses installed, the operation of the tester shall be checked at 1000, 500, 200, and 10 cfm standard air at each pressure condition of 12.5, 7.5, and 2.0 psig. All valves, gages, meters, adjustable louvers, accessories, and canopy seal system shall be checked for satisfactory performance during the test. All operating and service controls shall be readily accessible and easily operated by personnel wearing heavy gloves.

4.4.3 Calibration test. The 40 foot length of 4 inch supply hose and 0.5 inch static hose shall be connected to their respective fittings on the tester and connected to similar fittings on a test tank (160 cubic foot minimum volume). The output of the test tank shall be directed through a throttling valve into a measuring instrument specified in 4.3.3. A laboratory test gage with an accuracy of 0.50 of 1 percent of full scale shall be installed to read the pressure within the test tank. Other gages, manometers, and thermometers in accordance with 4.3.2 shall be installed as required. The calibration of the tester shall then be checked at 1000, 500, 200, 100, and 10 cfm standard air at each pressure condition of 10, 7.5, 6, 4.5, 3 psig. The airflow readings (in pounds per minute) of the tester (as determined by the testers, gages, flowmeter, conversion and correction charts) shall be within 3 percent or less of the readings of the calibration equipment at all pressure-flow settings listed. The accuracy of the pressure equalization indicator specified in 3.10.4 and the cabin pressure gage specified in 3.10.5 shall be checked against the test tank gage and a stopwatch (lowering and raising pressure in the test tank at the 5000 fpm rate).

4.4.4 Environmental test. The tester shall be subjected to the following tests. Upon completion of each environmental test, the test specimens shall be subjected to each of the tests specified in 4.4.2. Failure of the test specimen to operate as required during or after each exposure; failure of any item or assembly due to deterioration from exposures in this test; or formation of any undesirable condition that cannot be remedied by routine maintenance procedures shall be cause for rejection.

4.4.4.1 Low temperature. The tester shall be subjected to low temperature in accordance with the following procedure. The diesel engine testing requirements are specified in Appendix A, A.3.0. All gages, valves, meters, accessories, louvers and regulators shall be operated and shall perform satisfactorily in the -25°F ambient temperature condition.

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4.4.4.2 High temperature. The tester shall be subjected to high temperature in accordance with the following procedure. Upon completion of the 50 hour exposure, the ambient temperature shall be lowered to 120°F and the tests specified in 4.4.2 conducted. Failure of the aftercooler system to limit the temperature of the pressurized air leaving the test specimen to a maximum of 145°F while the test specimen is operating at full capacity of 1000 cfm standard air and 12.5 psi in the 120°F ambient shall be cause for rejection.

4.4.4.3 Rain test. The tester shall be subjected to rain (rain is rated at 4 inches of water per hour) in accordance with the following procedure. The tester shall be positioned on the test chamber floor in its normal operating position. After exposure to rain for 24 hours, the test specimen shall be started and operated at 1000 cfm and 12.5 psig for a period of 30 minutes. Failure of the test specimen to start or operate satisfactorily during the rain test shall be cause for rejection.

4.4.5 Endurance. The tester shall be connected through the 40 ft long, 4 inch ID, collapsible air hose specified in Par. 3.13 f to a throttling device to create the required pressure and flow conditions (pressures and flows shall be measured at the hose outlet) and the following endurance tests accomplished. Readings shall be recorded every hour for all gages, meters, and instruments of the test specimen and output flow, pressure. Barometric pressure and ambient temperatures shall be recorded. If 24 hour manning is not possible, the test periods may be fragmented, however, the data must be recorded for each test time period of operation to achieve the total number of hours in each case below:

a. Conduct the calibration tests specified in 4.4.3.

b. One hundred and ninety two hours of operation at 1000 cfm and 12.5 psi.

c. Forty two hours of operation at 500 cfm and 8 psi.

d. Forty two hours of operation at 200 cfm and 2 psi.

e. Forty two hours of operation at 200 cfm and 3.5 psi.

f. Forty two hours of operation at 100 cfm and 8 psi.

g. Forty two hours of operation at 10 cfm and 2 psi.

h. Conduct the calibration tests specified in 4.4.3.

NOTE: Comparison of this data in item h. with data recorded under item a. shall indicate whether a failure or undue wear has resulted from the endurance test.

4.4.6 Air compressor test. The canopy seal air compressor shall be operated at least 25 percent of the time during all endurance tests as specified in 4.4.5. The performance of the compressor shall be checked periodically during each test. Failure of the drive coupling or any marked decrease in performance or failure of a component part of the pressurizing system shall be cause for rejection.

4.4.7 Radio interference. The test specimen shall be tested as a complete unit for radio interference suppression. This test shall be conducted in accordance with MIL-STD-461 for radiated interference, except when a shielded room is not readily available a test location shall be selected in place of a

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shielded room. This shall be where the inherent background level and the area background level are within the acceptable limits specified in MIL-STD-461. The Testing Meter and test specimen shall have a common ground potential in substitution for the ground plane, and the meter antenna shall be a maximum of 20 feet from the perimeter of the test specimen.

4.4.8 Noise level. The tester noise level shall be measured at a maximum of 10 feet from the perimeter, and shall not exceed the values specified in 3.5.4. Measurements shall be taken out-of-doors on a paved surface in at least 12 different operator positions, 30° apart in a horizontal plan, parallel to the trailer bed. During this test, the test specimen shall be operated at full rated load with a 200 standard cubic feet per minute (SCFM) by-pass.

4.4.9 Operation in tilted position. The tester shall be placed on a platform with the parking brake(s) set to prevent it from rolling. The platform supporting the tester shall be tilted in eight different directions, 45° apart, on an 8.5° incline with the true horizontal plane. The test specimen shall be started and operated at 1000 cfm and 12.5 psi for 15 minutes in each position. The test specimen shall perform satisfactorily (4.4.2) or failure of a component (shutdown of the diesel engine) shall be a cause of rejection.

4.4.10 Serviceability. The test specimen shall be inspected and evaluated from the standpoint of ease of maintenance, servicing, operation and safety while maintenance personnel wearing heavy gloves or foul weather clothing. Particular note shall be made with regard to provisions made to prevent accumulation of snow, ice, or any element that may hinder servicing and operation.

4.4.11 Mobility. The test specimen shall be subjected to the applicable mobility tests that are the standard aerospace practices.

4.4.12 Safety demonstration. Test specimen shall be inspected and functionally checked to ascertain that the safety requirements have been incorporated in the design and manufacture.

4.4.13 Reliability testing. A test specimen shall be subjected to reliability testing in accordance with current, acceptable testing practices for a maximum MTBF of 500 hours. Test conditions shall be as specified under endurance with the testing described in 4.4.5.b representing one cycle to be repeated until an accept/reject decision is reached. Time accumulated under endurance testing may be counted toward reliability testing but in that case any failure experienced will also be used to reach an accept/reject decision.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of materiel is to be performed by DOD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Department or Defense Agency, or within the Military Department's System Command. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

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5.2 Marking. Interior and exterior containers shall be marked in accordance with industry practice. The nomenclature shall be as follows:

TESTER, PRESSURIZED CABIN LEAKAGE, LARGE AIRCRAFT
1000CFM, 12.5 PSIG
AF/M32T - 4 (Diesel Engine Driven)

6. NOTES

(This section contains information of a general or explanatory nature which may be helpful, but is not mandatory.)

6.1 Intended use. The AF/M32T- 4 tester covered by this specification is intended for use during ground checking and maintenance of large aircraft pressurized cabins. These items are unique to military applications of large wide body aircraft.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisitions documents should specify the following:

a. Title, number, and date of this specification.

b. Number of units to be tested (4.1.1).

c. Marking (5.2).

6.3 International agreement. Certain provisions of 3.13 of this specification are the subjects of international standardization agreement ABC Standard 17/33. When amendment, revision, or cancellation of this specification is proposed which will modify the international agreement concerned, the preparing activity will take appropriate action through international standardization channels including departmental standardization offices to change the agreement or make other appropriate accommodations.

6.4 Key Words

Altitude
Diesel engine
Endurance
Flowmeter
Interference
Noise
Rain
Storage
Towbar
Voltage

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APPENDIX

DIESEL ENGINE PERFORMANCE REQUIREMENTS

A.1 SCOPE

A.1.1 Scope. This document provides selection guidance and performance requirements for commercial diesel engines used in Air Force (AF) end item Support Equipment (SE). Unless otherwise specified, the term "engine" is synonymous with "diesel engine. This appendix is a mandatory part of the specification. The information contained herein in intended for compliance.

NOTE: This Performance Specification must only be used for End Items considered to be non-highway equipment. On-highway vehicles have different United States (US) Environment Protection Agency (EPA) emission requirements. On-highway vehicles would be self-propelled vehicles with typical on highway vehicle characteristics such as: forward and reverse drive, meet size and weight restriction, maintain highway speeds, designed for normal use over public roads, etc. All other equipment would be considered to be using non-road engines as described in this document including auxiliary engines installed in on-highway vehicles.

A.2 Applicable Documents

A.2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements documents cited in sections A.3 and A.4 of this specification appendix A, whether or not they are listed.

A.2.2 Government documents.

A.2.2.1 Specifications. The following specifications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

DEPARTMENT OF DEFENSE

MIL-L-2104 - Lubricating Oil, Internal Combustion Engine, Tactical service
MIL-T-5624 - Turbine Fuel, Aviation, Grades JP-4 and JP-5
MIL-M-38789 - Manuals, Technical, Overhaul Instructions With Illustrated Parts Breakdown
MIL-M-38807 - Manuals, Technical, Illustrated Parts Breakdown
MIL-L-46167 - Lubricating Oil, Internal Combustion Engine, Arctic Service
MIL-T-83133 - Turbine Fuels, Aviation, Kerosene Type

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FEDERAL

VV-F-800 - Fuel Oil, Diesel

STANDARDS

FEDERAL

FED-STD-5100 - Spark Arresters

(Unless otherwise indicated, copies of the above specifications and standards are available from the Standardization document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094).

A.2.3 Non-Government publications. The following document form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD are those listed in the issue of the DoDISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS are the same issues of the documents cited in the solicitation (see 6.2).

Society of Automotive Engineers (SAE)

SAE-J538 - Grounding for Storage Batteries (DoD adopted)
SAE-J541 - Voltage Drop for Starting Motor Circuits (DoD adopted)

(Application for copies should be addressed to the Society of Automotive Engineers, 400 Commonwealth Drive, Warrendale, PA 15096)

A.2.4 Order of precedence. In the event of a conflict between the text of this document appendix and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

A.3 ENGINE REQUIREMENTS

A.3.1 Engine assembly. The powerplant shall be diesel engine and be a self-contained unit without any external power source. The selection of a suitable diesel engine is the responsibility of the end item contractor.

A.3.2 Engine certifications. In addition to any other contract required engine certification or warranty, an engine manufacturer's certification must be obtained for any engine selected and included in the First Article Test Report or Bid Sample Test Report. The engine manufacturer must certify the following:

a. The engine selected by the end item contractor is acceptable for the intended use with the fuels and lubricants specified. The general engine application (cooling/starting etc.) shall also be approved.

b. The engine is a current production model and production is not scheduled to be terminated in the foreseeable future.

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c. At such time production is terminated, the engine manufacturer will, to the best of its ability, maintain spare parts support for a minimum of 10 years.

d. If the engine is not manufactured in the U.S., the manufacturer must have a complete distribution and service system in the U.S. for both commercial and Air Force requirements and the manufacturer must also certify compliance with the criteria outlined in Paragraph c above.

e. The engine shall satisfy all performance requirements set forth in the end item equipment specifications, this document, and all exhaust emission requirements as outlined in 2.3.7.

A.3.3 Basic engine requirements. The following basic requirements apply to any engine selected. The end item specification/description may tailor these requirements to fit particular end item requirements. For example, smaller lower power end items may use fewer sensors and shutdown equipment than larger higher-powered items.

A.3.3.1 Engine components and accessories. Engine components and accessories furnished with the engine shall be the engine manufacturer's most heavy-duty standard commercial items.

A.3.3.2 Identification marking. Unless otherwise specified, the engine shall be identified in accordance with engine manufacturer's standard commercial practice.

A.3.3.3 Engine rotation. Unless otherwise specified in the end item specification, the rotation of the engine shall be counterclockwise when viewed from the power takeoff end.

A.3.3.4 Emergency shutdown equipment. As a minimum, the engine shall be equipped with the following sensors, which shall be connected to an intake air or fuel solenoid shutdown device. Standard commercially available components shall be used.

a. Engine low oil pressure. This sensor shall activate when the oil pressure drops below 10 psi or below the minimum engine manufacturer's recommended pressure.

b. Engine overheat sensor. This sensor shall activate when engine temperature exceeds the engine manufacturer's recommended levels of coolant overheat, cylinder head overheat, or loss of coolant air in an air-cooled engine.

c. Engine overspeed sensor. This sensor will activate when the engine exceeds the manufacturer's maximum rpm or a critical end item system maximum rpm by 10%. This sensor is required if the engine exceeds 100 horsepower (hp) and optional if the engine is less than 100hp.

A.3.3.5 Engine sensors. The following sensors shall be provided to allow the operator to observe engine performance. Commercially available sensors will be used unless the end item specification stipulates the use of specific sensors.

a. Tachometer. This is optional when not required for system operation such as generator sets.

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b. Temperature. Cooling liquid temperature gauge or a suitable cylinder head gauge in the case of an air-cooled engine.

c. Liquid/air. Low liquid coolant level or loss of cooling air.

d. Oil. Oil pressure gauge.

e. Fuel. Fuel pressure gauge.

f. Water. The presence of water in water separator.

A.3.3.6 Engine Filters. Engine oil, air and fuel filters shall be supplied.

A.3.3.6.1 Sources. The engine oil, air and fuel filters shall be supplied by the engine manufacturer or its approved source. The filters shall meet the following requirements.

a. Availability. Be commercially available and of a standard type.

b. Heavy duty. Be designated by the engine manufacturer for the most severe service and the longest change interval.

c. Compatibility. Be compatible with the fuels and lubricants specified in 2.5.1 and 2.5.2, respectively.

A.3.3.6.2 Minimum Requirements. The individual filters shall meet the following minimum requirements.

a. Type. The air filter shall be a dry-type and provide maximum engine manufacturer's recommended filtration.

b. Filtration. The fuel and oil filter(s) shall provide filtration in accordance with the engine manufacturer's requirements.

c. Air. A cleanable clear prefilter shall be installed in the fuel supply system before the first engine filter so users can quickly determine if fuel contamination exists.

d. Fuel water separator. Water in the fuel shall be removed to less than 5 ppm at an efficiency of at least 99 per cent. A separate water separator or a combined fuel filter water separator may be used. The operator's engine control panel shall contain a water-in-fuel indicator.

A.3.3.7 Engine fuel injection system. Selection of a diesel engine and its associated fuel injection system shall be made with careful consideration given to the light kerosene turbine engine fuel requirements (See 2.5.1). To encourage use of available fuels, no engine changes are allowed when changing from conventional diesel fuel to light kerosene turbine fuels.

A.3.3.8 Engine controls. The diesel engine manufacturer shall provide a governor system suitable for the end item application.

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A.3.4 Engine performance requirements. The selected engine shall provide sufficient continuous power for the self-contained end item under all the conditions specified in the end item specification, using any of the fuel specified in 2.5.1.

A.3.4.1 Engine Power requirements. The required engine power shall be determined during the end item design process. The engine rated power shall be established by the emission requirements in 2.3.7. Special power ratings (other than standard commercial ratings) may be used but they must meet the emission requirements of 2.3.7. The selected engine should not be larger than 130 per cent of the worst case power required. Unless the engine supplier specifies differently, the maximum temperatures acceptable during any operating conditions specified are:

a. Oil Sump. Oil sump temperature 250°F.

b. Radiator. Radiator top tank (engine out) temperature 210°F.

A.3.4.2 Fuel consumption requirements. The selected engine shall demonstrate efficient fuel consumption while operating at full load on light kerosene turbine fuel. The following maximum full load fuel consumptions are noted:

a. Under 20 bhp - .500 lbs/bhp/hr

b. Over 20 bhp - .380 lbs/bhp/hr

A.3.5 Liquid cooled engine-cooling system. The cooling system shall be comprised of a radiator, radiator pressure cap, thermostat, fan, circulating pump, and interconnecting lines. The radiator shall be based on the engine supplier's guides. A pressure cap shall be used to protect the system from boiling during intermittent operation above the design conditions, reduce coolant loss from after-boil at shutdown, and ensure that a positive pressure exists at the water pump inlet. The engine shall be equipped with a 180°F thermostat unless the manufacturer recommends and warranties at another temperature. Cooling air management must be designed into the end item system so the heated cooling air does not stagnate in the end item.

A.3.5.1 Coolant. The coolant shall be a 60-40 per cent combination of commercial ethylene glycol antifreeze and water, respectively. The cooling system shall also be compatible with propylene glycol antifreeze.

A.3.5.2 Fan-Radiator. The fan and radiator shall be a standard commercially available system selected to meet the end item design requirements. The fan shall be equipped with guard assembly. The top tank temperature shall be limited to 210°F under all test conditions. The assembly shall not be excessively sized to meet this requirement. The radiator shall have a standard automotive type overflow container.

A.3.6 Air-Cooled engine cooling system. The selection of an air engine is not specifically prohibited and may be desired in some end items. The most significant problem with an air-cooled diesel engine may be in meeting the high altitude power requirements of the end item specification, and maintaining proper cooling airflow. If an air-cooled engine is selected, the complete blower and duct system shall be furnished by the engine manufacturer or his approved source. Cooling air management must be designed into the end item system so that the engine cooling air and engine combustion air is nearly the ambient air temperature and heated cooling air does not stagnate in the end item.

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A.3.7 Exhaust system. The exhaust system shall be designed to carry away combustion products from the engine. The design shall minimize restriction-to-flow to maintain a backpressure that will ensure proper engine life and operation. The exhaust backpressure at full load shall be a maximum of three inches Hg for naturally aspirated engines and two inches Hg for turbocharged engines. This pressure shall be measured at the exhaust manifold for naturally aspirated engines and after the turbine outlet for turbocharged engines.

A.3.7.1 Exhaust system installation. The exhaust system shall be designed to prevent produced heat from impairing the operation of adjacent components or destroying the protective finishes of components and the end item cabinet. It shall be designed for long life and be easy to disassemble during that life. The exhaust location shall be positioned to not interfere with the cooling air system intake, engine air intake, or operators' position. Exhaust system components shall be designed to withstand end item testing. The exhaust shall extend out side of the end item cabinet and shall be designed to prevent entrance of dirt, water, and debris.

A.3.7.2 Exhaust system materials. All components of the exhaust system tubing shall be fabricated from commercial type 409 or 304L stainless steel or other grades of low carbon stainless steel suitable for bending and/or welding with similar characteristics. Clamps, strips, hangers, etc. shall be fabricated from stainless steel, aluminum, aluminized steel, or mild steel properly treated for corrosion and high temperature resistance.

A.3.7.3 Spark arrester requirements. Spark arresters are designed to trap or pulverize a high percentage of glowing carbon particles which have been shown to have the ability to ignite cellulose material. The arresters do not make the end item explosion proof. Typically, end items will have many ignition sources in an explosive vapor atmosphere. If end item safety requires the use of spark arresters it shall meet the requirements of the U.S. Forest Service Standard 5100-1c. This standard specifically exempts turbo charged diesel engines when all the exhaust gasses pass through the pulverizing action of the turbine wheel. Spark arresters are not recommended for flight line equipment, but vehicles which might operate in dry grass or areas under the control of the National Park Service/Forest Service may need to temporarily or permanently meet the requirements of 5100-1. Spark arresters or spark arrester mufflers are not required unless specifically stated in the end item specification.

A.3.7.4 Muffler requirements. The muffler shall be a standard commercial design and provide noise control consistent with the overall unit noise requirements stated in the end item requirements without excessive backpressure. Fabrication from suitable stainless steel is preferred, but use of aluminized steel commonly used in the commercial muffler industry is acceptable.

A.3.8 Lubricating oil system. The selected engine shall provide a self-contained, full pressure lubrication system compatible with the oils specified in 2.5.2. A cooling liquid to engine oil heat exchanger shall be provided if required. In the case of air-cooled engines, a cooling air-to-oil heat exchanger shall be provided if required. The system shall be designed to limit lubricating oil temperature in the oil supply reservoir to 250°F maximum under worst case end item specified conditions.

A.3.8.1 Lubricating oil consumption. The selected engine shall not consume more than 0.0035 lbs/bhp/hr under the most severe operating conditions specified and measured at the conclusion of the First Article Testing.

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A.3.8.2 Lubricating oil level. The selected engine shall have a bayonet gage permanently marked to indicate "Full" and "Add". The gage shall be calibrated to indicate the proper level when the engine is secured. The low-level mark shall provide adequate engine lubrication under all operating conditions including tilted positions.

A.3.9 Engine exhaust and crankcase emissions. The Department of Defense (DOD) has been directed to exercise "Environmental Leadership" and meet all Federal, State, and Local environmental regulations. AF SE is designed to be mobile, portable, and standardized so forces from different locations can meet at a deployed site, and use common SE. Meeting local requirements with unique equipment is not practical so all SE must meet the strictest environmental requirements promulgated by the state of California or the EPA as outlined in the following paragraphs.

A.3.9.1 California air resources board (CARB), utility, lawn and garden engine (ULGE) rule. The CARB rule went into effect 1 January 1995 for engines rated 25 hp and below. In November 1998 the EPA Non-Road Engine Rule was revised to include all engine horsepower sizes. The revised Non-Road Engine Rule will preempt the CARB ULGE rule on 1 January 2000 when engines under 25 horsepower will have to meet the federal rule.

NOTE: Engines rated 25 hp or less must have CARB ULGE certification before production deliveries begin. This testing can be part of the bid sample testing required for Commercial Item Description (CID) end items or the first article test for performance specification end items. Engines should be selected that will meet the new EPA rule so that an engine change will not be required.

A.3.9.1.1 CARB ULGE. The CARB ULGE Rule is a complex legal/technical requirement document with many cross-references in California law. Engine manufacturers/importers normally have personnel with detailed knowledge of these requirements. Suggest end item contractors discuss these requirements with the engine suppliers. If additional information is required, including a list of engines that are certified (25 hp and below), contact Air Resources Board at 9528 Telstar Ave, El Monte CA 91731.

A.3.9.2 United States EPA, non-road engine rule (NER). On 17 June 1994, the EPA promulgated the NER for diesel engines 50 hp and greater rated power. In November 1998 the NER was revised to include all engine sizes and additional emission reductions over time. Actual emission standards will be phased in over time by horsepower classes for new engines as is shown in TABLE I.

A.3.9.2.1 Rule. The rule is complex, lengthy, and with many cross-references. Suggest end item contractors discuss this requirement with engine suppliers. If additional information is required, contact the U.S. EPA, Washington DC 20460; (202) 233-9256. The EPA does not maintain a list of engines meeting this requirement; individual suppliers maintain their own lists.

A.3.9.2.2 Engine selection. The engine selection process must take into consideration the changing emission requirements because the changing requirements may force the engine manufacturer to make significant engine changes. At worst case it may require the selection of another engine. This problem is especially critical for acquisition programs that will be spread over several years. Several steps may be taken to deal with this problem:

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TABLE I. Emission Standards in g/kW-hr (g/hp-hr)

Engine Power	Tier	Model Year	NMHC+ NOx	CO	PM
kW< 8 (hp<11)	Tier 1	2000	10.5 (7.8)	8.0 (6.0)	1.0 (0.75)
	Tier 2	2005	7.5 (5.6)	8.0 (6.0)	0.80 (0.60)
8<kW<19 (11<hp<25	Tier 1	2000	9.5 (7.1)	6.6 (4.9)	0.80 (0.60)
	Tier 2	2005	7.5 (5.6)	6.6 (4.9)	0.80 (0.60)
19<kW<37 (25<hp,50)	Tier 1	1999	9.5 (7.1)	5.5 (4.1)	0.80 (0.60)
	Tier 2	2004	7.5 (5.6)	5.5 (4.1)	0.60 (0.45)
37<75 (50<hp<100)	Tier 2	2004	7.5 (5.6)	5.0 (3.7)	0.40 (0.30)
	Tier 3	2008	4.7 (3.5)	5.0 (3.7)	-
75<kW<130 (100<hp<175)	Tier 2	2003	6.6 (4.9)	5.0 (3.7)	0.20 (0.15)
	Tier 3	2007	4.0 (3.0)	5.0 (3.7)	-
130<kW<225 (175<hp<300)	Tier 2	2003	6.6 (4.9)	3.5 (2.6)	0.20 (0.15)
	Tier 3	2006	4.0 (3.0)	3.5 (2.6)	-
225<kW<450 (300<hp<600)	Tier 2	2001	6.4 (4.8)	3.5 (2.6)	0.20 (0.15)
	Tier 3	2006	4.0 (3.0)	3.5 (2.6)	-
450<kW<560 600<hp<750	Tier 2	2002	6.4 (4.8)	3.5 (2.6)	0.20 (0.15)
	Tier 3	2006	4.0 (3.0)	3.5 (2.6)	-
kW>560 (hp>750)	Tier 2	2006	6.4 (4.8)	3.5 (2.6)	0.20 (0.15)

NOTE: NMHC + NOx is non-methane hydrocarbon + oxides of nitrogen. CO is carbon dioxide: PM is particulates

a. Select an engine that meets the most strict Tier III standards. While this is the most desirable and goes the farthest in fulfilling the direction to exercise "Environmental Leadership", it may not be practical because engines meeting Tier III may not be available until just before the Tier III standard goes into effect. At the very least an engine should be selected that the manufacturer can state the intention to carry through to Tier III with minimum changes so a completely new engine would not have to be selected.

b. Engines can be procured for the complete contract before they are no longer available. The NER prevents the engine manufacturers from stockpiling engines but there is no reason the end item contractor can't lay in the total contract requirements.

c. The engine manufacturer, with the procuring activity coordination, can ask the EPA for a National Security Exemption". This can only be used for valid military needs. An example would be for a few engines at the end of a long production run. The military need would be to keep the same engine for standardization, commonality, provisioning, spare parts, technical data, etc.

A.3.9.2.3 Crankcase ventilation kit. While gasoline engines have used positive crankcase ventilation from the early days of emission control, diesel manufacturers have been slow to accept the concept, especially on turbocharged and aftercooled diesels. Commercial crankcase ventilation kits are now

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available which filter and collect the crankcase vapors to minimize air pollution and eliminate condensation from dripping on the ground. The use of a crankcase vapor trap device is highly encouraged.

A.3.10 Accessory equipment. The following accessory equipment shall be provided as specified herein unless other specific requirements are included in the end item specification.

A.3.10.1 Power take-off. The engine shall be equipped with a commercially available power take-off components, shaft, chain, belts, etc. sized to the power and torque requirements of the end item and engine selected. As an option, a commercial power take-off clutch may be provided. The clutch shall be engaged during all normal operations, but may be disengaged for all cold starting or engine maintenance. The clutch shall be capable of being engaged with the engine operating at approximately 1000 rpm. The power take-off shall be designed to limit engine crankshaft radial and axial thrust loads to values less than manufacturer recommendations.

A.3.10.2 Fuel supply system. The fuel supply system shall be self-contained within the end item unless specified differently.

A.3.10.3 Fuel tank. End item fuel tanks must be manufactured to the highest standard to prevent tank failure from corrosion and/or fatigue cracking. Except as specified herein, standard commercial practice applies. The following requirements supplement standard commercial practice:

a. Capacity. Tank capacity shall support normal operations for 8 hours, unless, another time is specified.

b. Material. Metal tanks shall be made only with suitable commercial stainless steel or aluminum. Commercial 5052-H32/H34 aluminum is acceptable with typical welding and joint fabrication. Plain carbon steel shall not be used! Plastic tanks made from fiberglass, reinforced resin, or molded high-density polyurethane may be used with approval of the procuring activity.

c. Mounting. All tanks shall be mounted in a sturdy frame with band or strap holddowns. All cushioning or gasket material shall be nonporous, nonhygroscopic, weather resistant and fuel resistant. Brackets or tabs attached directly to the tank shall not be used.

d. Welding. Spot welds are not permitted under any condition. These welds create high stress concentrations that fail rapidly.

A.3.10.4 Fill opening and cap. The fuel tank shall have an external fill opening of not less than a 3 inch inside diameter and arranged to drain fuel spillage overboard. The fill opening cap shall be painted red and secured to the tank by a safety chain. The fill opening, cap, and chain shall be fabricated from non-sparking material.

A.3.10.5 Fuel gauge. A fuel gauge shall indicate the quantity of fuel in the tank with engine both operating and secured, but with the electrical system turned on. When a flange-mounted gauge sending unit is used, the mounting screws attaching the sending unit and the unit itself shall be made of corrosion resistant material.

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A.3.10.6 Fuel lines. All fuel lines shall be fabricated from non-corrosive metal and shall be disconnectable and easily removable for servicing. Soft solder or any other low-melting point material shall not be used. Where fuel lines are required to be flexible, coupled hose assemblies shall be used to eliminate using hose-clamp connections. A fuel tank shutoff valve shall be used to prevent continuous fuel spillage when lines are disconnected.

A.3.10.7 Fuel system materials. The fuel system sealers, elastomers, and plastics shall be compatible with all the fuels listed in 2.5.1.

A.3.10.8 Fuel system purge. The operator, at the engine control panel shall have a convenient way to quickly purge the air out of the fuel system when it has been run out of fuel. This purge system shall also provide air purge when the fuel system has air from maintenance actions.

A.3.10.9 Starting system. The engine shall be equipped with a starting system to meet the cold starting test requirements and any other requirements of this document or the end item specification. General starting system requirements are as follows:

a. Direct Current. DC electric starter motors with batteries, air starter motors with air tanks, and hydraulic starter motors with air charged accumulators are acceptable types of starting systems. Other innovative types may be proposed, but will have to be approved by the procuring activity.

b. Aids. External starting aids such as 110 volt plug in heaters are not allowed to meet the requirements of 3.2.1. Use of ether as a starting aid is not allowed. Self-contained aids such as fuel-fired heaters are acceptable.

c. Cold kits. Every end item product will have a complete starting system. Cold start kits are not permitted to meet the requirements of 3.2.1.

d. Drive. The starting system shall be approved by the engine manufacturer. The starter drive mechanism shall engage positively and release automatically.

e. Selection. A complete starting system design analysis shall be conducted to determine the best starting system to select and design. As a minimum, the analysis shall review 12 volt dc systems and predict starting performance, impact on end item design, cost, and overall reliability of the system.

A.3.11 Engine electrical system. A complete engine electrical system consists of a battery charging alternator, battery (or batteries), starter (if battery starting), and necessary wiring and relays. The system shall be approved by the engine manufacturer and be the heaviest duty usually offered for commercial service. If electrical shielding is required by the end item specification, it shall be incorporated into the electrical system in such a manner to provide access for maintenance action.

a. Voltage. Standard system voltage shall be 12 volts. If 24 volts are required for engine starting, it shall be provided by a 12 volt solenoid-connected series/parallel system.

b. Ground. All electrical systems shall be negative ground as noted in SAE Specification J538.

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c. Voltage Drop. The starter wiring system shall meet the voltage drop requirements of SAE-J541 for heavy-duty use.

A.3.11.1 Battery(s). All (starting and system) batteries shall be commercial maintenance free, sealed lead acid, starved electrolyte, gas recombination design meeting the requirements of MIL-B-18013/1. Other battery requirements are as follows:

a. Size. Batteries will lose a significant amount of energy from published values at -25°F and must be sized with a reduced capacity allowance.

b. Heater. Batteries will not readily take a charge at -25°F. Battery heating may be employed to aid charging, but battery overheat at high temperatures is not acceptable. Increased battery charging voltage may be employed using battery temperature compensated voltage regulation, but overcharging is not acceptable.

A.3.11.2 Battery compartment. The battery compartment shall be located in the vicinity of the engine starter and shall have adequate drains. The battery compartment shall be accessible to permit inspection of the battery (or batteries) without battery removal. The battery shall be located where the ambient temperature will not exceed excessive levels. The battery compartment shall be of rugged construction to withstand the weight of the battery (or batteries) under all conditions and tests specified for the end item.

A.3.11.3 Battery cables and terminals. Battery cables shall be in accordance with SAE-J541 for heavy duty use at the applicable voltage, except the positive terminal shall be colored red and labeled + and the negative terminal shall be colored black and labeled. Battery cables shall not be spliced.

A.3.11.4 Battery charging system. A commercial battery charging system with sufficient capacity shall be supplied.

A.3.12 Compressed air-starting system. A commercial diesel engine air starting system consisting of an air compressor, air storage tank, control valve, and air starter motor may be used. General comments are noted below:

a. Air starting would be an excellent choice if the end item already has an air compressor.

b. Air starters have the ability to deliver high power and thus ensure sufficient engine starting rpm.

c. A compressed air drier system such as is commonly used in the trucking industry is required to insure moisture freeze-up does not occur. Alcohol-type drier systems are not acceptable.

d. A readily accessible capped-off tap into the compressed air system is required for users to install a quick disconnect for recharging the air storage tank or jump starting from an external air system.

A.3.12.1 Assists to engine cold starting. Several types of engine accessories may be used to insure reliable cold starting. Ether starting aids are not permitted.

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a. Engine parasitic loads greatly increase engine starting system energy requirements in some end items. A conventional commercial friction clutch, an electric clutch, or some other type of mechanical device may be used to decouple the parasitic loads.

b. Ignition of engine fuel in the engine intake manifold has been used on diesel engines and a few systems are available commercially.

c. Several types of fuel fired heaters have been used, usually to heat the engine liquid coolant and the engine lubricating oil, but they may not be effective with short times allowed for the -25°F cold starting. Because diesel engines are very sensitive to the cylinder head, fuel, and combustion air temperature during cold starting it would be very effective to raise these temperatures during cold starting with a fuel-fired heater.

A.3.13 Human factors engineering. The engine, engine components, engine control panel, and engine accessories shall conform to accepted human factors engineering design criteria, as described in end item requirements or if not described, as used commercially.

A.3.14 Required fuels and lubricants. The selected engine shall operate satisfactorily under all conditions noted herein and in the end item specification for the fuels and lubricants listed below.

A.3.14.1 Engine fuels. The following shall be standard operating fuel unless noted:

a. DF2 per VV-F-800 (above 32°F)

b. DF-1 (or DF-A) Per VV-F-800 (below 60°F)

c. JP-5 per MIL-T-5624 (all environments)

d. JP-8 per MIL-T-83133 (all environments)

NOTE: A new additive called +100 (or JP-8+100) is being used at some AF facilities which may have some undesirable effects on diesel fuel filters and water separators. The issues are under study and because it is strictly a military additive, no commercial data exists. For the purpose of engine selection, JP-8+100 is not considered an operational or emergency fuel.

e. Jet A and Jet A-1 commercial aviation turbine engines (all environments) Jet A will be standard First Article Test Fuel.

A.3.14.2 Engine lubricant. The engine utilized shall be compatible with Grade 15W40 of MIL-L-2104 from 0°F, to the highest temperatures specified for the end item. For temperatures below 0°F, the oil will be preheated before engine operation. The engine also shall be compatible with Arctic oil per MIL-L-46167 from -40°F to +60°F and to limited use to +100°F in Arctic environments where seasonal oil changes are not practical. Seasonal oil changes may be required in most locations.

A.3.15 Engine vibration analysis. For most end items the engine will be the main source of vibration energy. Vibration energy has long been recognized as a major cause of reduced reliability and increased maintenance on powered end items. A vibration analysis, including both longitudinal and torsional

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vibrations shall be conducted. Longitudinal vibrations shall be measured in the x, y, and z plains to determine the maximum vibrations. The engine and/or the engine subassembly shall have maximum vibration deflection of .020 inch (peak to peak). The engine and/or engine subassembly shall be mounted so as to reduce vibrations transferred to the main end item by a 4 to 1 reduction factor. Subassemblies, enclosures, control boxes, control panel, fenders etc. shall have less than .040 inch displacement.

A.4 VERIFICATION

A.4.1 Engine testing requirements.

A.4.1.1 Environmental tests. The following tests are required to determine the selected engine and accessory equipment suitability for Air Force worldwide operations. These tests shall be accomplished with the engine installed in the end item. The tests may be conducted concurrently with the other end item tests or as separate tests. The test schedule is a generic flight line SE schedule and may be changed with end item user coordination and strong evidence that the particular end item will see a significantly different schedule in actual operation.

A.4.1.2 Cold starting. The engine shall demonstrate cold start capability using the following procedures. Units properly winterized with suitable fuel, lubricants, and hydraulic fluid shall have unlimited stand-alone starting and restarting capability at -25°F. They shall have operational capability, as specified in the end item design, usually -40°F, but the starting system performance (time to start and operate or the requirements to use outside aids) may degrade below -25°F. Stand-alone means no starting aids may be used that are not contained within the units (i.e., no outside aids such as 110 volt power or separate heater, etc.). With temperatures between 0 and -25°F, the starting system shall allow the unit to start and be ready to perform any function in its basic design within five minutes of operator start initiation. With the temperature above 0°F, the unit shall be ready to perform any function within one minute of operator start initiation. The starting system energy storage subsystem shall have the capacity at -25°F to provide a minimum of two starts. The starting energy replenishment subsystem shall at -25°F, recover from one start in one hour running time and two starts in two hours running time. Unless specifically deleted by other contract documentation, this test is required even if no end item cold testing is required including bid sample testing.

A.4.1.2.1 Cold starting test conditions. The following test conditions shall be maintained. The cold start test shall be accomplished with the engine installed in the end item.

a. Engine fuel. The fuel shall be a light kerosene, such as JP-5, JP-8, Jet A or Jet A1; commercial No. 1 Diesel; DF-1 or DF-A per VV-F-800; with cloud point below -25°F.

b. Engine lubricating oil. The engine lubricating oil shall be drained, flushed and filled with arctic lubricating oil per MIL-L-46167.

c. Hydraulic system. The hydraulic system, if used, shall be winterized.

d. Starting system precondition for air start systems. Immediately before placing the end item in the cold chamber, the starting system shall be cycled a minimum of five times in an ambient condition with a

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minimum of 70 per cent relative humidity. The starting system energy storage shall be at full capacity when installed in the chamber.

e. Temperatures. The critical temperatures shall be at or below -25°F before a start is attempted. In order to reduce cold chamber test time, and if chamber conditions permit, the chamber may be lowered below test temperatures, and/or forced cooling used to return critical temperatures to required values. Removable components such as batteries may be removed and placed in the direct cold air stream to accelerate their return to test temperatures.

A.4.2 Critical temperatures. As a minimum, the following critical temperatures shall be recorded and shall be at or below the required test temperature.

a. Ambient air. Measured and averaged in four locations approximately equal distance from the end item and the chamber walls and three feet off the ground. The average chamber temperature shall not exceed 0°F during any of the run periods.

b. Energy storage device. Measured from the approximate center of mass of the device (i.e., battery electrolyte, hydraulic accumulator fluid, air tank, etc.).

c. Engine lubricating oil sump

d. System hydraulic fluid tank (if used)

e. Engine air intake system

f. Engine coolant or cylinder head for an air-cooled engine

g. Fuel tank fuel

h. Any other location that directly would affect starting performance

A.4.2.1 Test procedures.

a. Critical temperatures at or below -25°F:

(1) Start within five minutes and run five minutes.

(2) Shutdown and re-stabilize critical temperatures.

(3) Start within five minutes and run two hours.

(4) Shutdown and re-stabilize critical temperatures.

(5) Start within five minutes and run for one hour.

(6) Shutdown and re-stabilize critical temperatures.

(7) Start within five minutes and run for one hour.

NOTE: The starting energy storage shall be at 90% or above of capacity by the end of the evaluation.

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b. Test Failures. Failure to start, to start within the times allowed, failure to recharge, and failures other than those which can be corrected with simple hand tools (such as loose connection, leaking line, etc.) would constitute test failure and require re-testing.

c. Test data recording. In addition to the critical temperatures and as a minimum the engine cranking rpm and the starting energy re-charging shall be recorded.

d. Test Locations. Selection of test locations is the contractor responsibility. The cold starting test facility should be selected carefully to insure it has sufficient size and refrigeration capacity. Previous Air Force cold testing was accomplished at the all-weather room of the McKinley Climatic Laboratory, Eglin Air Force Base, Florida (904) 822-4087. White Sands Missile Range also has an excellent temperature test facility (505) 678-3201.

A.4.3 High temperature and high altitude test. The engine shall be monitored for compliance with the requirements of 2.3.2 during specified end item testing.

A.5 PACKAGING

(This section is not applicable to this appendix.)

A.6 Notes

(This section contains information of a general or explanatory nature which may be helpful, but is not mandatory.)

A.6.1 Engine performance requirements and analysis. The test report requirements shall be identified in the procurement contract and shall include a section dealing with the end item horsepower requirements and the performance of the engine utilized. As a minimum the following information shall be included:

a. Analysis of the end items horsepower requirements.

b. The engine manufacturer's data on the engine utilized, the rationale for the selection, and the engine manufacturer's warranty information.

c. Analysis of the engine performance during the first article test including an analysis of the cooling air management requirements.

d. Engine starting system analysis.

e. Engine manufacturer's certification as required in 1.3.

A.6.2 Alternate engine technical order (T.O.) requirements for support equipment engines.

A.6.2.1 T.O. requirements. A separate engine T.O. overhaul and repair instruction with IPB is required and this requirement shall be identified in the procurement contract. This T.O. must follow TM-86-01- guidelines provided by SA-ALC/TILTR and must be in sufficient detail to permit complete and total overhaul in such a manner as to allow the engine to function to the same performance and reliability levels as when new. Full and complete instructions for field level inspection and maintenance must be included, preferably as a separate section.

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a. Show only the engine configuration(s) used in the particular end items(s). If the same engine is used in several different end items, the engine T.O. will clearly show each application and engine configuration.

b. Instructions and artwork should meet the general requirements of MIL-M-38789A and MIL-M-38807A. They can be obtained from SA-ALC/TILTR, Bldg. 171, 485 Quentin Roosevelt Rd., Ste #2, Kelly AFB TX 78241-6421. Specific format requirements are waived in the interest of cost reduction. Data which is suitably clear and concise may be extracted directly from directly from commercial data.

c. The Illustrated Parts Breakout, (IPB), listing shall be in MIL-M-38807A format with Figure/Index No., Part No., FSCM, description, units per assembly, and usable on Code. The SMR code can be either in the MPL or preferably in the numerical index which is also required. Individual figure index numbers do not have to begin with 1, providing the numbering system is not confusing. The manufacturer must also include a Provisioning Parts List, (PPL), in the format required by the acquiring agency.

d. Operation and routine service information shall be included in the basic equipment operation manual.
e. A draft manual shall be approved by SA-ALC/LDPG, Kelly Air Force Base, Texas 78241 before formal Air Force acceptance.

f. Commercial T.O. data is unacceptable due to the following:

(1) Often written in confusing multiple languages.

(2) Usually shows many different engine configurations, which are very confusing to the users.

(3) The IPB breakout listing does not show manufacturers' codes, SMR codes, use on codes, and often does not show the complete component P/N.

(4) Commercial manual cannot be updated readily when needed.

CUSTODIAN:
Air Force – 99

PREPARING ACTIVITY:
Air Force - 82

Project Number – 4920-0925

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.

2. The submitter of this form must complete blocks 4, 5, 6, and 7, and send to preparing activity.

3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	**1. DOCUMENT NUMBER** MIL-PRF-32051	**2. DOCUMENT DATE (YYYYMMDD)** 19991112

3. DOCUMENT TITLE **TESTER, PRESSURIZED CABIN LEAKAGE, LARGE AIRCRAFT, TRAILER MOUNTED**

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME *(Last, First, Middle Initial)*	b. ORGANIZATION	
c. ADDRESS *(Include Zip Code)*	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON *(if applicable)*	**7.DATE SUBMITTED** (YYYYMMDD)

8. PREPARING ACTIVITY

a. NAME SA-ALC/TILDD	b. TELEPHONE *Include Area Code)* (1) Commercial (210) 925-6314 (2) AUTOVON 945-6314
c. ADDRESS *(Include Zip Code)* 485 QUINTEN ROOSEVELT ROAD KELLY AFB, TX 78241-6425	**IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT:** Defense Standardization Program Office (DLSC-LM) 8725 John J. Kingman road, Suite 2533, Ft. Belvoir, VA 22060-2533 Telephone (703) 767-6888 AUTOVON 427-6888

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